



Faculty of Science, Engineering and Built Environment

SIT112 Introduction to Data Science and Artificial Intelligence

Deakin University Unit Guide

Trimester 2, 2026

CONTENTS

Welcome	2
Who is the unit team?	2
Unit chair: leads the teaching team and is responsible for overall delivery of this unit	2
Other members of the team and how to contact them	2
Help with administrative and academic queries	2
About this unit	2
Unit development in response to student feedback	2
Learning outcomes	3
Assessing your achievement of the unit learning outcomes	3
Hurdle requirements	3
Summative assessment task 1	4
Your learning experiences in this unit	5
Educator-facilitated (scheduled) learning activities - on-campus unit enrolment	5
Educator-facilitated (scheduled) learning activities - online unit enrolment	5
Typical study commitment	5
Learning activities	6
Unit learning resources and equipment	6
Bring Your Own Device (BYOD)	7
Essential learning resources	7
Recommended learning resources	7
Key dates for this study period	7
Unit weekly activities	7

Welcome

Data science is an emerging field and data scientists must be able to know how to make sense of data. In SIT112, students will develop knowledge of fundamentals in data science, in particular data manipulation and algorithms for analytics. The unit will also cover the practice of data science including ethical and responsible behaviour when crawling, cleaning, analysing, representing and repurposing the data. Students will be able to obtain data, recognise data formats, summarise and visualise relationships in the data, perform exploratory data analysis tasks and build predictive models. This Unit Guide provides you with the key information about this unit. Please read it carefully and refer to it frequently throughout the study period. Your unit site also provides information about your rights and responsibilities. We will assume you have read this before the unit commences, and we expect you to refer to it throughout the study period.

To be successful in this unit, you must:

- read all materials in preparation for your learning activities and follow up each with further study and research on the topic
- start your assessment tasks well ahead of the due date
- read or listen to all feedback carefully and use it in your future work
- attend and engage in all educator facilitated (scheduled) learning activities and other learning experiences as part of the unit design

Who is the unit team?

Unit chair: leads the teaching team and is responsible for overall delivery of this unit

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Campus: Melbourne Burwood

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Other members of the team and how to contact them

Details of any other staff will be provided on the unit site at the start of the trimester.

Help with administrative and academic queries

If you have any concerns about the unit during the trimester, help is available from a range of sources. It's best to seek support early to discuss your concerns as adjustments can be made if appropriate. Support is also available as follows:

- SEBE Student Hub section in your unit site – for faculty-specific support
- Unit Chair or Campus Leader – for academic matters such as content, assessments, and learning support
- [Student Central](#) – for administrative or enrolment-related queries (e.g., course advice, special consideration, or fee queries), or to make an appointment with a student adviser.

For additional support information, please see the 'Your Rights and Responsibilities' section under 'Content' in your unit site.

About this unit

Unit development in response to student feedback

Every trimester, we ask students to tell us, through eVALUate, what helped and hindered their learning in each unit. You are strongly encouraged to provide constructive feedback for this unit when eVALUate opens (you will be emailed a link). In previous versions of this unit, students have told us that these aspects of the unit have helped them to achieve the learning outcomes:

- The use of generative AI tools such as ChatGPT was encouraged.
- The coherency of the lectures, workshops, and tasks led to a practical understanding of the contents.
- Inquiries received quick attention from the teaching team.
- Acquiring data analysis skills through Python was identified a valuable achievement, offering substantial benefits to

students pursuing careers.

- Engaging in tasks proved to be an enjoyable and stimulating experience.

The following aspects of the unit have been introduced, enhanced or retained in response to feedback from students who have undertaken this unit in previous trimesters:

- Measures have been introduced to improve the feedback process and online engagement.
- Workshops will emphasize human aspects of AI.
- The EoUA has been removed.
- The size of certain assessment tasks has been reduced to allow for more in depth engagement.
- More targeted programming assistance will be provided to the students in the workshops.
- We will continue to ensure that the level of feedback provided by markers for tasks is adequate.
- We continue to embrace a responsible use of external resources such as ChatGPT to achieve the objectives of the unit.

Learning outcomes

Each unit in your course is a building block towards Deakin's Graduate Learning Outcomes - not all units develop and assess every Graduate Learning Outcome (GLO).

ULO	These are the Unit Learning Outcomes (ULOs) for this unit. At the completion of this unit, successful students can:	Alignment to Deakin Graduate Learning Outcomes (GLOs)
ULO1	Describe key concepts, algorithms, components of data science, and artificial intelligence and their applications in real-world scenarios.	GLO1: Discipline-specific knowledge and capabilities
ULO2	Describe the synergies between data science and artificial intelligence, and outline the connections between these fields.	GLO1: Discipline-specific knowledge and capabilities
ULO3	Adapt and explain programming scripts that make use of common data science and artificial intelligence algorithms, techniques, and tools.	GLO1: Discipline-specific knowledge and capabilities GLO3: Digital literacy
ULO4	Effectively communicate artificial intelligence and data science solution designs and resulting insights, to technical and non-technical stakeholders.	GLO2: Communication
ULO5	Evaluate the social and ethical implications of artificial intelligence adoption.	GLO1: Discipline-specific knowledge and capabilities GLO8: Global citizenship

Assessing your achievement of the unit learning outcomes

Hurdle requirements

To be eligible to obtain a pass in this unit, students must meet certain milestones as part of the Learning portfolio.

Summary of the hurdle requirement/s	Rationale
1. Unit Tasks (Learning Portfolio) Students need to complete the unit's pass tasks to pass the unit. The portfolio requires students to work through a series of Tasks to produce a portfolio of evidence showing achievement of the unit learning outcomes. To achieve a passing grade, students must demonstrate adequate performance in the completion of the unit's pass tasks, which will show the required minimum standard. Credit, Distinction, and High Distinction tasks are used to determine higher grades, corresponding to achievement of the unit learning outcomes to a higher level.	The pass tasks in this unit provide students the opportunity to develop and demonstrate achievement of the Unit Learning Outcomes at the minimum expected standards. These tasks are included as hurdle requirements so that students are able to provide evidence of achievement of these ULOs through their portfolio. The portfolio that they submit is used to measure their performance against the minimum standards as well as their ability to justify the outcomes that they have achieved through self-assessment and reflection. The hurdle requirement also provides a mechanism for student-staff interaction to check progress and address educational and motivational issues before it is too late in the trimester.
2. Task Discussion Students are required to respond to feedback via OnTrack comments, in a timely manner, and answer questions posed by the teaching team. For some tasks, the teaching team may require an audio or video recording, in-class discussion, in-person meeting, or online meeting to be used to demonstrate student understanding of the task and its associated concepts. In these cases, any video recordings or online meetings must clearly show the student talking through the concepts and responding to the questions posed by the teaching team.	Task discussions help students to demonstrate achievement of their unit learning outcomes, while also helping to authenticate student learning and improve academic integrity in the unit. The student-staff interaction during discussions will allow the teaching team to make judgements about student learning and progress in select tasks that are representative of high-order thinking and learning. Feedback resulting from this will help aid student learning and provide them with additional opportunities to demonstrate achievement of unit learning outcomes.

Summative assessment (tasks that will be graded or marked)

It is your responsibility to keep a backup copy of every assignment and the materials used to develop/complete it where possible (e.g. written/digital reports, essays, videos, images). In the unusual event that one of your submissions becomes corrupted, is incorrectly submitted or otherwise lost, you may be asked to submit the backup copy. Any work you submit may be checked by electronic or other means for the purposes of detecting breaches of academic integrity such as collusion, plagiarism and contract cheating. You must understand your responsibility to act with honesty and integrity in your studies as Deakin takes all breaches very seriously. Make sure you read [Your rights and responsibilities as a student in this unit](#) to find out more about academic integrity.

Deakin has a universal assessment submission time of 8 pm AEDT/AEST. A late penalty will apply to assessments submitted after 11.59 pm AEDT/AEST. Australian Eastern Standard Time (AEST) is observed from the first Sunday in April to the first Sunday in October, while Australian Eastern Daylight Time (AEDT) is observed from the first Sunday in October to the first Sunday in April, during daylight saving time in all states apart from Queensland, NT and WA, which remain as AEST.

Summative assessment task 1

Title	Learning Portfolio
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Brief description of assessment task	In this unit, assessment is designed to encourage and reward students for demonstrating achievement of the unit learning outcomes; with higher grades representing better achievement of these outcomes. The unit employs OnTrack, a web application designed specifically to support learning and assessment activities. Working regularly and completing the tasks on time will help students collect the necessary evidence for their portfolios. The learning portfolio will include work associated with the tasks that are to be submitted, revised, and discussed with the teaching team during the trimester.
Individual/Group	Individual
Detail of student output	This is an individual assessment task in which students will produce a variety of artefacts to demonstrate their comprehension of the unit. The tasks will require students to apply programming skills in order to comprehend and create solutions for data gathering, cleaning, preparation, visualization, and analysis as well as making predictions. The evaluation of the tasks will be based on the accuracy and thoroughness of the solutions, as well as the student's capability to explain and refine their solutions when necessary.
Grading and weighting (% total mark for unit)	100% - marked and graded
This task assesses your achievement of these Unit Learning Outcome(s)	ULO1 – through describing and working with these concepts, algorithms, and components. ULO2 – through indicating connections between the concepts and fields presented. ULO3 – through adapting and explaining programming scripts. ULO4 – through communicating findings and solutions. ULO5 – through reflecting upon ethical implications.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1 – through exploring concepts, algorithms and components of data science and artificial intelligence and working with scripts to demonstrate these. GLO2 – by communicating resulting solutions and insights from projects undertaken. GLO3 – through student ability to use data science and artificial intelligence scripting tools. GLO8 – through reflecting upon the broader impact of data science and artificial intelligence.
How and when you will receive feedback on your work	Students will be required to work on and submit tasks for assessment and feedback each week. The teaching team will then check the task's solution to provide individual feedback through the OnTrack submission system to assist the student in completing tasks and achieving the target grade for the unit.
When and how to submit your work	At the end of the unit, students will use OnTrack to combine together the artefacts they have created and a learning summary report into a single portfolio for assessment by Friday 2 October 2026, 8pm AEST (Week 12) .

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Your learning experiences in this unit

Educator-facilitated (scheduled) learning activities - on-campus unit enrolment

1 x 2 hour online lecture per week, 1 x 2 hour practical experiences (workshop) per week, weekly meetings.

Educator-facilitated (scheduled) learning activities - online unit enrolment

Online independent and collaborative learning including 1 x 2 hour online lecture per week (recordings provided), 1 x 2 hour online practical experiences (workshop) per week, weekly meetings.

Typical study commitment

Students will on average spend 150 hours over the teaching period undertaking the teaching, learning and assessment activities for this unit.

This will include educator guided online learning activities within the unit site.

For campus students the learning and assessment activities will include online lecture time as described above, weekly

meetings (previously drop-in sessions), designated activities in the practical workshops, assessment tasks, readings and study time. For online students the activities will include online lecture time as described above, weekly meetings (previously drop-in sessions), designated activities in the online practical sessions, discussion boards, assessment tasks, readings and study time.

There are many resources available to you while undertaking your study in this unit. Learning opportunities are offered via lectures, practical sessions and the unit site. Students should begin by reading the unit information, the intended unit learning outcomes and assessment activities. Lecture notes and reading materials provide an overview of the content for each week. Students should review this information in preparation for engaging in discussions during lectures, practical sessions and consultation sessions.

Students should also conduct further research by examining information on the web, supplementary material in recommended readings and textbooks. Attending lectures and designated consultation sessions regularly will enable students to focus on key concepts and engage in discussions with the teaching staff. Lecture recordings will also be available for review via the unit site. Students must participate in the discussion forums, engage in learning with other students, and monitor the forum and respond regularly and appropriately.

Learning activities

At Deakin, courses are delivered within a learning environment that provides all students with equitable and consistent access to facilities, infrastructure, resources, and support to assist student progress and achievement of learning outcomes.

‘Lectures’ are the activities where teaching staff engage you through presentations with student participation.

In ‘seminars’, an educator will guide you in a smaller group of students through highly interactive discussions and activities.

Your units may also include ‘practical experiences’ such as ‘laboratory’, ‘workshops’, ‘clinical skills’ and more. These hands-on activities typically take place in specialised facilities with industry tools, equipment or technology to allow you to apply your knowledge practically.

If you see a ‘meeting’ in your timetable, this is an optional drop-in session.

‘Assessments’ or ‘team-based learning’ indicate an evaluation of your skills or knowledge. A ‘pre-assessment practice’ could be scheduled to prepare you for these assessments. Visit the [About CloudDeakin – Assessment webpage](#), or the [Understanding Assessments webpage](#) for more information.

Take a look at the [Learning activities webpage](#) for a full list of learning activity types.

Generative AI

Deakin welcomes the opportunity to engage with emerging technologies in education and seeks to build your capability in the ethical and responsible use of current and emergent technology. Deakin also upholds a commitment to academic integrity and to ensuring high-quality educational outcomes that prepare you for an AI-driven future.

For more information on the acceptable use of GenAI, [visit Using GenAI at Deakin](#), read your assessment instructions and talk to your unit chair.

Unit learning resources and equipment

Your unit learning resources can be accessed from your unit site.

The texts and reading list for [SIT112](#) can be found via the University Library.

Note: Select the relevant trimester reading list. Please note that a future teaching period's reading list may not be available until a month prior to the start of that teaching period so you may wish to use the relevant trimester's prior year reading list as a guide only.

The [Deakin Software Library](#) provides students with access to software that you may need or find useful for your study at

Deakin.

Bring Your Own Device (BYOD)

To fully engage with Deakin's learning experiences, students must be able to access and use internet-connected devices as outlined in [computing requirements](#) at Deakin.

To support student success at Deakin, we have a bring-your-own-device (BYOD) learning environment that acknowledges that students and educators bring with them the digital tools they regularly use to complete academic tasks. These tools stay with you beyond the classroom, helping you to keep learning, explore ideas more deeply, and connect with knowledge in ways that matter to you.

Students requiring a loan device should visit our [Loan Laptop](#) webpage or students requiring longer-term assistance should visit our [Student Financial Assistance](#) webpage.

Essential learning resources

There is no prescribed textbook for this unit.

Recommended learning resources

Due to the interdisciplinary nature of data science, the following books will be recommended and are available in the University Library:

- **Data Science from Scratch: First Principles with Python**, Joel Grus, O'Reilly Media, 2019.
- **Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython**, Wes McKinney, O'Reilly Media, 3rd edition, 2022.
- **Bird's Engineering mathematics**, John Bird, 9th edition, 2021.

Additional recommended readings are also available in the Library and include:

- **Python Data Science Handbook: Essential Tools for Working with Data**, Jake Vanderplas, O'Reilly Media, 2023.
- **Murach's Python for Data Analysis**, Scott McCoy, Mike Murach & Associates, Incorporated, 2021.
- **Intro to Python for Computer Science and Data Science: Learning to Program with AI, Big Data and The Cloud**, Paul Deitel, Harvey Deitel, Pearson Education Limited, 2020.

Key dates for this study period

You will find all important dates for your 2026 enrolment, including teaching periods, holidays, census dates, withdrawal dates, exam periods and deadlines to apply to graduate in the [2026 Handbook – Dates and deadlines](#)

Trimester 2 teaching period begins	Monday 6 July 2026
Census date	Saturday 15 August 2026
Intra-trimester break	Monday 10 August - Friday 14 August 2026
Trimester 2 teaching period ends	Friday 25 September 2026
Study period (end-of-unit assessment/ examination preparation period)	Monday 28 September - Friday 2 October 2026
End-of-unit assessment/examination period	Monday 5 October - Friday 16 October 2026
Inter-trimester break (the period between trimesters)	Monday 19 October - Friday 30 October 2026
Unit results released	Friday 30 October 2026 (10.30am)

Unit weekly activities

Week	Commencing	Topic	Assessment due date
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1	6 July 2026	Introduction to Data Science and AI	
2	13 July 2026	Essentials for Data Analysis	
3	20 July 2026	Data Visualization	
4	27 July 2026	Getting Data	
5^	3 August 2026	Cleaning Data	
6	17 August 2026	Preparing Data	
7	24 August 2026	Data Analysis	
8	31 August 2026	Time Series Analysis	
9	7 September 2026	Making Predictions	
10	14 September 2026	Natural Language Processing	
11#	21 September 2026	Human-centred AI	
12 - Study period	28 September-2 October 2026		Learning portfolio due 2 October
End-of-unit assessment and exam period	5-16 October 2026		

^ Intra-trimester break: **Monday 10 August - Friday 14 August 2026** (between weeks 5 and 6)

AFL Grand Final Eve public holiday: **Friday 25 September 2026** - University closed